Discovering, Establishing, and Designing for Requirements: An HCI Case Study

Undergraduate CHI-Style Assignment Template

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Abstract

This report presents a complete requirements-to-conceptual-design account for [system or service]. It documents how the team scoped a design problem, gathered and analyzed evidence, established defensible requirements, and translated those requirements into a conceptual design. Replace bracketed text with a coherent account of one project; do not leave the abstract as a list of activities. The finished abstract should state the target users, setting, methods, principal findings, requirements, and design contribution in 150–200 words.

Keywords

human-centered design; requirements; data gathering; data analysis; conceptual design; HCI education

1 Introduction

1.1 Problem, users, and setting

[Name the setting] currently requires [primary users] to [goal], but this is difficult because [observed or evidenced breakdown]. This project investigates how an interactive system could support [bounded outcome] without assuming that a new application is automatically the right answer. The unit of analysis is [individual / team / service encounter], and the study concerns [location, frequency, constraints, and stakeholders]. State what is known from preliminary observation, prior research, or institutional context; clearly label conjecture as conjecture.

The project is grounded in a human-centered process: discover the problem space, gather evidence about people and practice, interpret that evidence, negotiate requirements, and develop a conceptual design [5]. This sequence is iterative, not a claim that requirements become permanently fixed after one study.

1.2 Research questions and contribution

The study asks: **RQ1**: How do [users] currently accomplish [goal]? **RQ2**: Which needs, constraints, and opportunities should shape a future design? **RQ3**: How can a conceptual design make the proposed interaction understandable and accountable? The contribution is a documented, evidence-traceable requirements set and concept for [system], not merely a polished interface.

Expected length. 1.5–2 pages, including the context figure and Table 1.

Common mistakes. Opening with a solution; treating all people as "the user"; claiming needs without evidence; using research questions that can be answered only by opinions about the proposed interface.

Table 1: Project scope. Replace all bracketed entries.

Element	Project decision
Primary users	[Who directly uses or is affected by the system?]
Stakeholders	[Who administers, supports, regulates, or is indirectly affected?]
Activity	[Observable task or practice under study]
Setting	[Physical, social, organizational, and technical context]
Boundary	[What is intentionally outside this assignment?]

Placeholder figure

A context photograph, service blueprint excerpt, or stakeholder map for [setting]. Remove identifying information.

Replace with a legible, original study artifact.

Figure 1: [Replace this caption.] Describe what the figure shows and why it matters.

Writing guidance

Write in the past tense for completed study work and present tense for stable context. Give one concrete example of the existing workflow. Replace every bracketed placeholder, and cite sources for contextual claims.

Instructor note

Assess scope discipline: a strong introduction identifies a feasible population and activity, distinguishes stakeholders, and makes a testable design opportunity visible.

2 Discovering Requirements

2.1 Stakeholders, assumptions, and problem space

Requirements discovery identifies the people, activities, values, and constraints that define a design space [5]. We began with a stakeholder inventory. Primary participants were [group]; secondary stakeholders included [group]; and gatekeepers or domain experts included [group]. Their interests were not assumed to align: for example, [participant group] values [value], whereas [organization] must also satisfy [policy, safety, cost, or privacy constraint].

Table 2: Stakeholder analysis.

Stakeholder	Relationship	Interest, influence, and concern	
Primary user			
	Direct use	[Goal; capability; risk if the system fails]Secondary user>	Secondary user
	Indirect use	[Coordination need or downstream effect]Gatekeeper>	Gatekeeper
	Access/control	[Permission, policy, resource, or ethical concern]Domain expert>	Domain expert
	Knowledge source	[Practice knowledge and potential bias]	

Placeholder figure

A stakeholder map or current-state journey. Use arrows to show relationships and label tensions, handoffs, and pain points.

Replace with a legible, original study artifact.

Figure 2: [Replace this caption.] Describe what the figure shows and why it matters.

We recorded initial assumptions in a risk register: [assumption 1], [assumption 2], and [assumption 3]. Each assumption was marked as either evidence-backed, uncertain, or contradicted after fieldwork. This protects the project from designing around the team's own habits rather than participants' practices.

2.2 Current practice and opportunity areas

Describe a representative journey from trigger to outcome. For [persona role], the activity begins when [trigger]. They then [step], coordinate with [person or tool], and finish when [criterion]. Friction appears at [breakdown], where [consequence]. Do not convert every complaint into a feature request: distinguish a cause, an observed workaround, and a possible design opportunity.

Three preliminary opportunity areas were [opportunity A], [opportunity B], and [opportunity C]. We carried these forward as questions to investigate, rather than commitments to implement. Contextual inquiry is useful when work is situated and tacit, because it lets a researcher ask about decisions close to the activity [1].

Expected length. 2–2.5 pages. Include a stakeholder table and one current-practice artifact.

Table 3: Data-gathering protocol and evidence produced.

Stage	Researcher action	Evidence and safeguard
Recruitment	[Invite and screen]	[Eligibility; voluntary participation]
Opening	[Consent and context]	[Information sheet; permission to record]
Core activity	[Observe/interview/task]	[Notes, quotes, artifacts]
Closing	[Debrief and compensate]	[Withdrawal/contact information]
After session	[De-identify and store]	[Pseudonym; access control; retention]

Common mistakes. Listing demographics instead of stakeholders' roles; treating a persona as data; saying "users need an app"; confusing a desired feature with a human need; omitting organizational and accessibility constraints.

Writing guidance

Use short, concrete vignettes: "When ..., [role] ... because ...". Mark statements as observation, participant account, document evidence, or team inference. Show why the problem is consequential for people, not just technically interesting.

Instructor note

Look for a credible problem-space account. The best work exposes tensions and uncertainty, then uses research to resolve them rather than defending an early solution.

3 Data Gathering

3.1 Method rationale and study design

We selected [interviews / observations / survey / diary study / document analysis] because it can reveal [kind of evidence] in [context]. The method complements rather than replaces [second method]: [method A] captures [strength], while [method B] checks or broadens [strength]. Data-gathering choices should be driven by questions, practical constraints, and participants' wellbeing [5].

Participants were recruited through [channel] using [inclusion criteria]. We sought variation in [experience, role, access need, or context], not statistical representativeness. State the final sample honestly: n = [ ], with [distribution]. Explain any attrition or access limitation and how it affects the claims that follow.

3.2 Procedure, ethics, and data management

Each session lasted approximately [duration]. After consent, participants [procedure steps]. We gathered [audio, notes, artifacts, survey responses] and stored them in [approved location] with [pseudonyms/access controls]. Participation was voluntary; participants could skip questions or withdraw [state applicable policy]. Avoid collecting identifiers that the analysis does not require.

The protocol included open prompts such as "Tell me about the last time you [activity]" and follow-ups such as "What happened next?" Questions about actual episodes reduce speculative

Placeholder figure

A data-gathering timeline: recruitment, consent, session activities, debrief, and secure handling of data.

Replace with a legible, original study artifact.

Figure 3: [Replace this caption.] Describe what the figure shows and why it matters.

answers. Pilot [number] session(s) revealed [change], so we revised [prompt/order/material].

Expected length. 2.5–3 pages. Include enough protocol detail for a peer to reproduce the study responsibly.

Common mistakes. Claiming a convenience sample represents everyone; asking leading questions; describing consent as a signature only; failing to state what data were recorded; reporting a survey without its questions, scale, recruitment, or response count.

Writing guidance

Write the procedure chronologically. Separate what you planned from what happened. Include one or two actual question stems, but place the full guide in Appendix A.

Instructor note

Grade the fit between research questions and method. Ethical care, sampling transparency, and a plausible procedure matter more than an unnecessarily large sample.

4 Data Analysis, Interpretation and Presentation

4.1 Preparation and analytic approach

We analyzed [data types] using [thematic analysis / content analysis / descriptive statistics / affinity diagramming]. First, [transcription, cleaning, or familiarization step] prepared the corpus. We then assigned concise codes to meaningful segments, compared codes across participants and contexts, and grouped related codes into candidate themes. Themes were reviewed against raw evidence and renamed to express an interpretive pattern, not a topic label [2].

For transparency, each finding below has an evidence trail: source material → code → theme → design implication. [Number] team member(s) coded the material. Explain how disagreements were discussed, how a codebook changed, and how negative cases were treated. Do not report inter-rater reliability unless your analysis design actually used it and you can explain it.

4.2 Findings

Theme 1: [interpretive statement]. Participants described [pattern]. For example, P[ID] said, "[short, anonymized quotation]."

Placeholder figure

An affinity map excerpt, coded journey, or chart. Show enough labels to make the transformation from data to theme understandable.

Replace with a legible, original study artifact.

Figure 4: [Replace this caption.] Describe what the figure shows and why it matters.

This matters because [interpretation tied to context], not merely because the quote is vivid.

Theme 2: [interpretive statement]. Evidence from [participant/artifact/observation] showed [pattern], including the counterexample [case]. This suggests [implication] only within [boundary].

Theme 3: [interpretive statement]. [Explain pattern, variation, and consequence.] Quantitative results, if used, should give denominator, measure, and uncertainty where appropriate: "[x]/[n] respondents selected [response]" is more informative than "most users agreed."

Expected length. 3–3.5 pages. This is often the intellectual center of the report.

Common mistakes. Presenting quotes as self-explanatory; turning every code into a theme; erasing contradictory evidence; reporting percentages with a tiny or unstated denominator; mixing result and proposed feature in the same sentence.

Writing guidance

Use descriptive headings that make an argument ("Coordination breaks down at handoffs"), not headings such as "Interview question 3." Present evidence first, interpretation second, and implication third. Retain a de-identified audit trail in Appendix B.

Instructor note

Strong analysis is traceable and selective. It explains why patterns matter to participants' activity, acknowledges variation, and avoids claiming universality from a small qualitative sample.

5 Establishing Requirements

5.1 From findings to requirements

Requirements state what a system and its surrounding service must enable or respect; they should not prematurely prescribe a screen layout [5]. We translated each supported finding into a requirement by naming the actor, condition, action or quality, and success criterion. Requirements were checked with [participants, stakeholders,

Table 4: Evidence-to-interpretation traceability matrix. Use anonymized identifiers.

Evidence source	Code / observation	Theme	Interpretation and bounded implication
P01 interview, [time]	[Concise segment meaning]	[Interpretive theme]	[Why this pattern matters; do not write a feature yet]
Observation [site], [time]	[Action/workaround/breakdown]	[Theme]	[Contextual explanation and exception]
Artifact [ID]	[Documented constraint]	[Theme]	[Constraint on future requirement]

Placeholder figure

A requirements traceability diagram linking findings, requirement IDs, and later conceptual-design elements.

Replace with a legible, original study artifact.

Figure 5: [Replace this caption.] Describe what the figure shows and why it matters.

expert, or literature] through [walkthrough, prioritization activity, or review].

Functional requirements describe supported actions. Data requirements identify the information needed, its source, ownership, sensitivity, and lifecycle. Non-functional requirements express qualities such as accessibility, privacy, reliability, learnability, and performance. Organizational requirements describe policy, staffing, integration, or governance conditions. The four categories can interact: an apparently simple reminder feature may create data, consent, and notification constraints.

5.2 Prioritization and acceptance criteria

We prioritized requirements using [MoSCoW / value-risk matrix / stakeholder workshop]. A "Must" item is essential for the defined release or study scenario, not merely popular. Priority reflects evidence, consequence of absence, implementation dependencies, and equity impacts. For each high-priority requirement, specify a testable acceptance criterion: e.g., "Given [condition], [user] can [action] with [measure] without [harm]."

Requirements remain provisional when evidence is thin. R[ID] is flagged for validation because [uncertainty]. This honest uncertainty is a useful design result: it directs the next study rather than disguising a decision as fact.

Expected length. 2.5–3 pages, including a requirements table with at least 8–12 substantive entries.

Common mistakes. Writing features instead of needs; using vague terms such as "easy" without a criterion; prioritizing by personal preference; omitting privacy and accessibility; presenting all requirements as equally important; failing to trace requirements to evidence.

Writing guidance

Use active, testable wording: "The system shall enable [role] to ..." or "[Role] needs to ... when ...". Include the rationale and evidence ID beside every important requirement. Do not invent requirements merely to fill categories.

Instructor note

Assess requirement quality, not quantity. A good table supports later design decisions, identifies trade-offs, and makes it possible to challenge an unsupported item.

6 Conceptual Design

6.1 Design vision and conceptual model

The conceptual design proposes [system name], a [type of interactive product/service] that helps [role] [goal] in [context]. Its central design principle is [principle], derived from R[ID] and T[ID]. The system model is [object/action model]: users work with [key objects] and can [key actions]. Define these terms in language participants recognize; a conceptual model should help users predict what actions are possible and what outcomes follow [4].

The proposed interaction is [direct manipulation / conversation / form fill-in / notification / embodied interaction / hybrid]. Explain why this style fits the activity, environment, devices, attention demands, and access needs. State deliberate exclusions, such as "The system does not make [high-stakes decision]; it supports [human role] with [information or coordination]."

6.2 Scenario and task flow

Scenario: [name]. [Persona/role] begins when [trigger]. They see [information] and choose [action]. The system then [response], making [status, consequence, or uncertainty] visible. If [exception], the system [recovery or escalation]. The scenario ends when [success criterion]. Annotate each step with the requirement(s) it serves.

6.3 Design alternatives and trade-offs

We considered [alternative A] and [alternative B]. Alternative A was rejected or deferred because [evidence-based trade-off]; alternative B was retained because [reason]. Discuss effects on workload, privacy, accessibility, control, error recovery, and organizational practice. Participatory design is especially valuable when those affected can shape consequential design choices [3].

Expected length. 3–3.5 pages. Include a model/flow and a requirement-to-design mapping.

Table 5: Requirements catalogue. Add rows; retain stable IDs for traceability.

ID	Type	Requirement	Evidence / rationale	Priority
R1	Functional	The system shall enable [role] to [action] when [condition].	Theme T[ID]; P[ID]	Must
R2	Data	The system shall collect/display [data] only for [purpose] and [retention rule].	Constraint [ID]	Must
R3	Quality	A [user with access need] shall complete [task] using [support/criterion].	Observation [ID]	Should
R4	Organizational	[Responsible role] shall be able to [govern/support action].	Stakeholder review	Could

Table 6: Requirement-to-conceptual-design mapping.

RequiremenConceptual-design response		How it will be checked
R1	[Object, action, and feedback]	[Scenario / usability task]
R2	[Data boundary or consent mechanism]	[Policy review / walkthrough]
R3	[Accessibility or resilience support]	[Inclusive test condition]
R4	[Administration or service process]	[Stakeholder review]

Placeholder figure

A conceptual model diagram or task flow. Use named objects, actions, feedback, and exception paths; annotate with R1–R[n].

Replace with a legible, original study artifact.

Figure 6: [Replace this caption.] Describe what the figure shows and why it matters.

Common mistakes. Jumping directly to annotated screens; calling a feature list a conceptual model; ignoring exceptions; treating automation as neutral; using a persona without linking it to study evidence; failing to explain why an interaction style fits the context.

Writing guidance

Describe the interaction before drawing it. Name objects consistently across scenario, diagram, and prototype. A conceptual design may use low-fidelity sketches, but each sketch must answer a requirement or reveal a decision.

Instructor note

Look for a design that is intelligible before visual polish. Strong submissions show traceability, explicit trade-offs, and appropriate limits on what the technology decides.

Table 7: Evaluation plan tied to requirements.

Task / R-ID	Method and evidence	Decision rule
	Task	
, R1	[Think-aloud; completion, errors, quote]	[Revise Task if...]Task>
, R2	[Walkthrough; data-flow understanding]	[Escalate Task if...]Task>
, R3	[Inclusive session; barrier observed]	[Revise if...]

7 Evaluation Plan

7.1 Evaluation questions and methods

The next iteration will evaluate whether the conceptual design supports the priority requirements, rather than whether participants simply like it. Evaluation questions are: (1) Can [role] understand the model and complete [task]? (2) Does [feature/flow] create confusion, workload, exclusion, or privacy concern? (3) Does the design fit [organizational/workflow constraint]? A moderated think-aloud walkthrough with [target participants] will test comprehension and task flow; a stakeholder review will test operational feasibility.

The prototype will be [paper / clickable low-fidelity / service enactment], intentionally matched to the questions. A paper prototype can test sequence, labels, and feedback but cannot establish real-world reliability. State what the prototype cannot validly evaluate.

7.2 Success, risk, and iteration

Success criteria derive from Table 5. For example, [x]/[n] participants should [task outcome], and no participant should encounter an unaddressed [severity] breakdown related to R[ID]. Qualitative observations will be coded for misunderstandings, workarounds, breakdowns, and unanticipated value. Findings will lead to [revision rule], while high-risk concerns such as [privacy/safety/equity concern] pause implementation until reviewed.

Expected length. 1.5–2 pages. This is a planned study: use future tense and avoid inventing results.

Common mistakes. Using satisfaction as the only outcome; testing only happy paths; claiming a prototype tests what it cannot; writing success measures unrelated to requirements; overlooking participant risk or accessibility in the evaluation.

Placeholder figure

An evaluation journey showing participant tasks, observation points, measures, and revision decisions.

Replace with a legible, original study artifact.

Figure 7: [Replace this caption.] Describe what the figure shows and why it matters.

Placeholder figure

A concise roadmap from the completed study to the next validation, prototype iteration, and stakeholder decision.

Replace with a legible, original study artifact.

Figure 8: [Replace this caption.] Describe what the figure shows and why it matters.

Writing guidance

For each task, specify the starting condition, prompt, expected outcome, evidence collected, and decision triggered by a failure. Avoid a generic “we will conduct usability testing” sentence.

Instructor note

Reward a realistic, requirement-driven evaluation plan. Its purpose is to reduce uncertainty and guide revision, not to certify a design as finished.

8 Discussion and Conclusion

8.1 What the process established

This project established that [main supported finding] shapes the design of [system] because [reason]. The resulting requirements prioritize [two or three priorities], and the conceptual design responds through [named design moves]. The value of the work lies in the chain from situated evidence to a design that can be evaluated, rather than in claiming that one small study has discovered universal user needs.

8.2 Limitations and next steps

The study is limited by [sample, setting, duration, access, method, or researcher-position limitation]. Consequently, findings should be interpreted as [bounded claim]. The next iteration should [specific evidence-gathering or evaluation action], particularly to examine [uncertain requirement or underserved stakeholder]. Reflect on how the team's position, access, and assumptions may have influenced what participants shared and how evidence was interpreted.

Table 8: Bounded claims and next steps.

Current claim or uncertainty	**Next evidence or design action**	
What the present study supports		
	[What design decision itWhat remains uncertain informs]What remains uncertain>	What remains uncertain
	[Specific participant group, method, or review]Risk or limitation>	Risk or limitation
	[Safeguard and decision owner]	

8.3 Conclusion

Conclude in one paragraph by restating the problem, evidence-based contribution, and immediate next design decision. Do not introduce new claims here. The final sentence should make clear what a future designer or researcher can do differently because of this report.

Expected length. 1–1.5 pages.

Common mistakes. Repeating the abstract word-for-word; claiming proof from a small study; hiding limitations; proposing a new feature in the conclusion; ending with a generic statement that technology will improve life.

Writing guidance

Use this section to make a bounded argument. Name both the contribution and its limits. A candid limitation increases credibility when it is connected to a concrete next step.

Instructor note

The conclusion should demonstrate reflective HCI practice: evidence, consequence, uncertainty, and a thoughtful path for iteration.

References

[1] Hugh Beyer and Karen Holtzblatt. 1998. Contextual Design: Defining Customer-Centered Systems. In *Proceedings of CHI*. ACM, 109–110.

[2] Virginia Braun and Victoria Clarke. 2006. *Using Thematic Analysis in Psychology*. Vol. 3. Qualitative Research in Psychology. 77–101 pages.

[3] Michael J. Muller. 1993. Participatory Design: The Third Space in HCI. In *Human Factors in Computing Systems*. ACM, 31–32.

[4] Donald A. Norman. 2013. *The Design of Everyday Things* (revised and expanded ed.). MIT Press.

[5] Jennifer Preece, Yvonne Rogers, and Helen Sharp. 2023. *Interaction Design: Beyond Human–Computer Interaction* (6 ed.). John Wiley & Sons.

A Study Materials

A.1 Participant information and consent checklist

Before beginning, provide: the study purpose; what participation involves; duration; recordings and data handling; foreseeable discomforts; benefits, if any; voluntary participation and withdrawal process; contact details; and the applicable approval statement. Do not include a participant's name, signature, contact information, or other identifying data in this submitted report.

A.2 Semi-structured interview guide

(1) Tell me about the last time you [performed the focal activity]. What happened from beginning to end?
(2) What parts were easy, difficult, slow, risky, or frustrating? Can you show me an example?
(3) Which people, tools, documents, and information did you use? What happened when one was unavailable?
(4) What workarounds have you developed, and what do they cost you or others?
(5) If a new system supported this activity, what would it need to respect about your practice, privacy, access, or control?
(6) Is there anything important we did not ask? Which of your answers should we treat cautiously?

Interviewer reminders. Ask neutral follow-ups ("Can you say more?"); do not promise a feature; do not pressure participants to disclose sensitive information; record analytic memos separately from factual notes.

Expected length. 1–2 pages, only as needed. This appendix supports the method; it does not replace the method description.

Common mistakes. Including identifiable data; presenting leading questions; omitting a withdrawal/debrief procedure; pasting a raw transcript without purpose or permission.

Instructor note

Check that the guide could plausibly generate the evidence used in the Findings section and that it treats participants with respect.

B Analysis Audit Trail

Reflexive memo. [Write 100–200 words on how the researcher's role, expectations, relationship to the setting, language, or analytic decisions may have shaped this account. State one action taken to examine that influence.]

Expected length. 1–2 pages. Include representative evidence, not every note.

Common mistakes. Treating codes as objective facts; exposing identifiable quotations; claiming triangulation merely because multiple data types exist; omitting a negative case or analytic uncertainty.

Instructor note

An audit trail makes reasoning inspectable. It is not a demand for mechanical objectivity; it shows disciplined interpretation.

Table 9: Example de-identified analytic audit trail.

Source ID	Excerpt or observation	Initial code	Theme decision and memo
P01	"[Short anonymized excerpt.]"	[Code]	[Why it supports, revises, or challenges Theme T1]
P03	[De-identified field-note excerpt]	[Code]	[Comparison with P01; negative case if applica-Artifact ID ble]Artifact ID>
	[Relevant non-sensitive content]	[Code]	[Constraint or implication]